Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 71%. We keep your highest score.

Next item →

Skill progress from this assignment Beta

Prototyping and Strategic Evaluation of Generative AI +15 XP

251/500 XP

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1. In a scenario where Dana wants to convert an LLM's response into a CSV format using LangChain, which output parser should they use? 1 / 1 point

- ○ Dana should use the PandaDataFrameParser for this task.
- ◉ Dana should use the CommaSeparatedListOutputParser.
- ○ Dana should use the XMLParser for this task.
- ○ Dana should use the JSONOutputParser for this task.

✓ Nice work

Dana should use the CommaSeparatedListOutputParser to convert an LLM's response into CSV format in LangChain.

2. Maria is implementing a chain using LangChain Expression Language (LCEL). What is the correct sequence of steps to create a functional LCEL pattern? 1 / 1 point

- ○ Invoke with input values → Define a template with variables → Create a PromptTemplate → Build a chain with the pipe operator
- ○ Create a chain with the pipe operator → Define variables → Build a PromptTemplate → Invoke with input values
- ○ Create a PromptTemplate → Build a chain with the pipe operator → Define a template with variables → Invoke with input values
- ◉ Define a template with variables → Create a PromptTemplate → Build a chain using the pipe operator → Invoke with input values

✓ Nice work

This represents the proper sequence for creating an LCEL pattern, starting with defining the template structure and ending with invoking the completed chain.

3. In LangChain, how does an agent integrate with external tools to fulfill user requests? 1 / 1 point

- ○ By running predefined commands without external integration.
- ○ By storing user inputs in memory for later execution.
- ◉ By using the language model to determine actions and then querying databases or websites.
- ○ By creating a new tool for each user request.

✓ Nice work

Agents use the language model to determine actions and then integrate with tools such as databases or websites to fulfill requests.

4. When working with long documents in LangChain, why is the text splitting process important? 1 / 1 point

- ◉ To break documents into chunks that fit within model context windows
- ○ To improve the aesthetic appearance of text in the application
- ○ To eliminate duplicate content across documents
- ○ To reduce token count and associated API costs

✓ Nice work

The primary purpose of text splitting is to divide long documents into smaller chunks that can fit within the limited context window of language models.

5. What is the purpose of the FewShotPromptTemplatein LangChain? 1 / 1 point

- ○ To store conversation history for future reference.
- ◉ To provide specific examples or shots for LLMs, guiding the model to generate the requested output.
- ○ To visualize data using natural language.
- ○ To execute LLM outputs directly without examples.

✓ Nice work

The FewShotPromptTemplate in LangChain provides specific examples or shots for LLMs, guiding the model to generate the requested output.

6. How does LangChain use memory in its applications? 1 / 1 point

- ○ By storing only the final output of each chain.
- ○ Using sophisticated technical algorithms
- ○ Physically altering devices to gain access
- ◉ By reading from and writing to memory to ensure continuity across interactions.

✓ Nice work

LangChain uses memory to read and write historical data, ensuring continuity and context preservation across interactions.

7. Nathan is migrating from traditional LangChain SequentialChain to LCEL. Which syntax represents the correct way to connect components in LCEL? 1 / 1 point

- ◉ chain = prompt_template | llm | output_parser
- ○ chain = [prompt_template, llm, output_parser]
- ○ chain = prompt_template.connect(llm).connect(output_parser)
- ○ chain = prompt_template >> llm >> output_parser

✓ Nice work

LCEL uses the pipe (|) operator to connect components in a chain, allowing for a more intuitive and functional programming approach.